



A Discontinuous Galerkin Finite Element Method for Spherical Advection on Octahedral Meshes

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1. INTRODUCTION

Standard strong-form nodal discontinuous Galerkin (DG) methods on curved spherical manifolds often suffer from severe **geometric aliasing**. Non-polynomial rational metric terms interact with the polynomial basis, violating the Discrete Geometric Conservation Law (DGCL) and causing artificial mass drift during long-term climate simulations.

To resolve this issue, this work introduces a novel weak-form differentiation matrix satisfying the **Summation-by-Parts (SBP)** property. Exact mass conservation is enforced at the linear algebra level, completely bypassing geometric aliasing.

2. MATHEMATICAL FORMULATION

We consider scalar advection on a 2D spherical manifold $\Omega \subset \mathbb{R}^3$:

$$\frac{\partial q}{\partial t} + \nabla_s \cdot (\mathbf{v}q) = 0 \quad \text{on } \Omega$$

where $q(\mathbf{x}, t)$ is the scalar field and $\mathbf{v}(\mathbf{x})$ is a given non-divergent tangent velocity field corresponding to steady solid-body rotation.

3. MESH MAPPING & FIELD SETUP

Numerical integration on triangular elements employs symmetric quadrature node distributions corresponding to polynomial degrees $k = 1 \sim 4$:

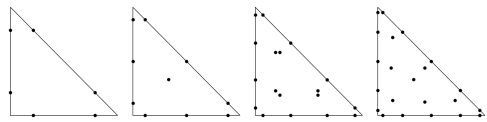


Figure 1: Triangular Element Quadrature Nodes ($k = 1 \sim 4$)

Octahedral Projection: Base octahedron faces are mapped onto the unit sphere ($\mathbf{x}_s = R \frac{\mathbf{x}_{\text{flat}}}{\|\mathbf{x}_{\text{flat}}\|}$).

Initial Field $q_0(\mathbf{x})$

3D Gaussian bell centered at $\mathbf{x}_c = (1, 0, 0)^T$ ($\sigma = 10.0$):

$$q_0(\mathbf{x}) = \exp(-\sigma r^2)$$

where $r = \arccos(\mathbf{x} \cdot \mathbf{x}_c)$.

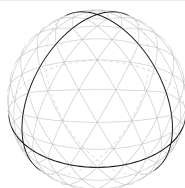


Figure 2: Spherical Mesh

Wind Field $\mathbf{u}(\mathbf{x})$

Solid-body rotation driven by $\Omega = \omega(\sin \alpha_0, 0, \cos \alpha_0)^T$:

$$\mathbf{u} = \Omega \times \mathbf{x} = \begin{pmatrix} -\omega y \cos \alpha_0 \\ \omega x \cos \alpha_0 - \omega z \sin \alpha_0 \\ \omega y \sin \alpha_0 \end{pmatrix}$$

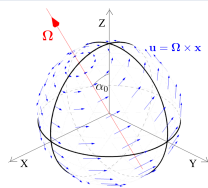


Figure 3: Tilted Wind Field

4. SBP-PRESERVING DIFFERENTIATION MATRIX

To ensure strict discrete integration-by-parts on curved meshes with non-linear Jacobian metric terms, we construct a modified differentiation operator D_η^{new} satisfying the SBP property:

$$D_\eta^{\text{new}} = V_\eta V^T W + \frac{1}{2} (W^{-1} + VV^T) B (I - VV^T W)$$

where $V^T W V = I$ represents the orthonormality relation under diagonal mass/weight matrix W , and B is the boundary flux matrix.

5. NUMERICAL RESULTS

Advection simulations performed for $t = 12$ days (1 full revolution) using polynomial order $k = 4$ with a fourth-order five-stage low-storage Runge-Kutta (LSRK4) scheme.

Error Evaluation Metrics:

L^2 Error Norm

$$e_{L^2} = \sqrt{\frac{\int_\Omega (q_{\text{ex}} - q_h)^2 d\Omega}{\int_\Omega q_{\text{ex}}^2 d\Omega}}$$

L^∞ Error Norm

$$e_{L^\infty} = \frac{\max |q_{\text{ex}} - q_h|}{\max |q_{\text{ex}}|}$$

Relative Mass Error

$$\Delta M_{\text{rel}} = \frac{|\int_\Omega (q_h - q_0) d\Omega|}{\int_\Omega q_0 d\Omega}$$

Table 4.1: Convergence Rates at $t = 12$ Days ($k = 4$)

N	L^2 Error	L^∞ Error	L^2 Order	L^∞ Order
4	5.683e-03	2.787e-02	—	—
8	3.185e-04	2.820e-03	4.2	3.3
16	1.516e-05	1.635e-04	4.4	4.1
32	7.379e-07	9.423e-06	4.4	4.1

L^2 & L^∞ Error Evolution Across Grid Resolutions

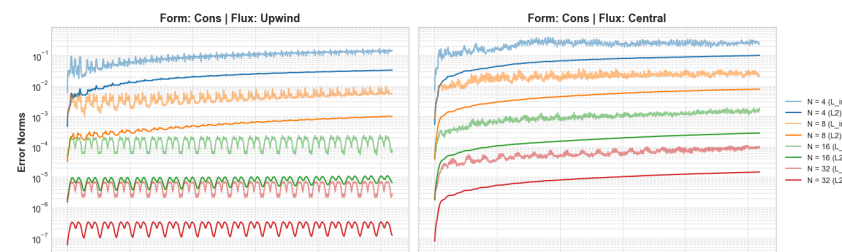


Figure 4: L^2 and L^∞ error evolution over 12 days across grid resolutions ($N = 4 \sim 32$).

Mass Error Evolution Across Grid Resolutions

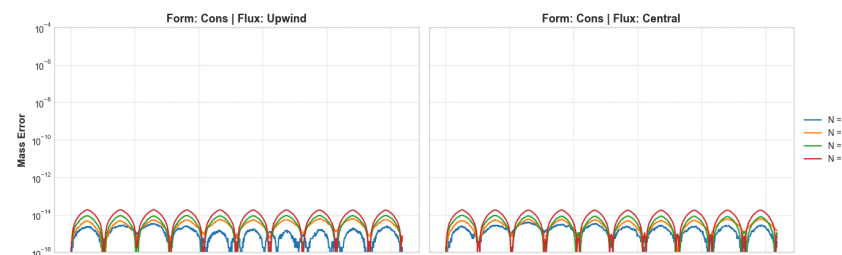


Figure 5: Relative mass error tracked over 12 days, bounded within $\pm 10^{-14}$.

6. CONCLUSION

- Developed an SBP-preserving DG method for spherical advection on Octahedral meshes.
- Embedded boundary topology and mass matrices directly into discrete derivative operators (D_η^{new}), ensuring discrete integration-by-parts.
- Achieved optimal high-order convergence while maintaining **exact discrete mass conservation down to machine precision** ($\pm 10^{-14}$).

7. BIBLIOGRAPHY

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